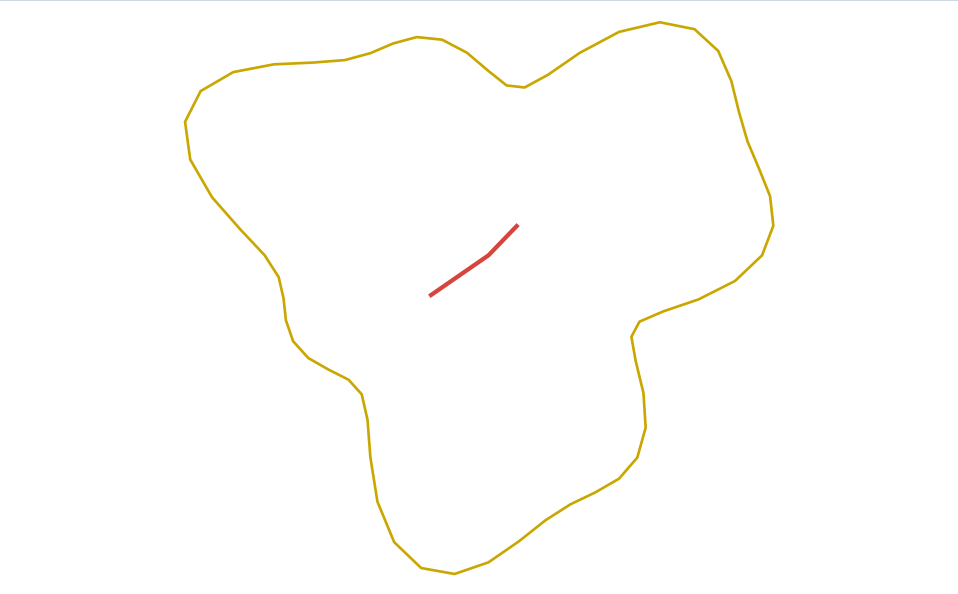
# DEEP Detailed Assessment

## Eastern Corn Belt Plains reference assessment

### *Mink Brook*

| | |
|---|---|
| Source | StreamCurves reference-curve development |
| Version / status | v6 · Preliminary |
| Predictor source | streamcat |
| Region match | Not resolved |
| Coordinates | 43.68581234, -72.23671234 |
| COMID / HUC8 | 9327042 / 01080104 |
| Drainage area | 12.43 km2 |
| HR reach watershed area | 11.98 km2 |
| Reach length | 1000.0 ft |
| Watershed basis | HR reach watershed (STAF site engine v0.5.0) |
| Desktop values | STAF site engine (HR reach watershed) 5, StreamCat lookup engine (NHDPlus V2 basin) 2 |
| StreamCat reach | COMID 9327042 (this reach) |
| Content digest | sha256:bc2948a42062bb90030672afe31eb3271f563c790dee145c5cb8d7af8dfa42fd |
| Ecosystem Condition Index | 0.45 to 0.82, condition not determinable, 4 of 20 functions not assessed |
| STAF function coverage | 16 of 20 (4 documented exclusions) |
| Flagged transfers | none |

### *Watershed*



Map background unavailable

### *Function scores (0-15)*

| Function | Score | Condition |
|---|---|---|
| Catchment hydrology | 12.0 | F |
| Surface water storage | | Not scored |
| Reach inflow | 5.9 | AR |
| Streamflow regime | 8.4 | AR |
| Low flow and baseflow dynamics | 8.8 | AR |
| High flow dynamics | 9.5 | AR |
| Floodplain connectivity | 14.9 | F |
| Hyporheic connectivity | | Not scored |
| Channel evolution | 14.9 | F |
| Channel and floodplain dynamics | 15.0 | F |
| Sediment continuity | | Not scored |
| Bed composition and bedform dynamics | | Not scored |
| Light & thermal regime | 7.3 | AR |
| Carbon processing | | Not scored |
| Habitat provision | | Not scored |
| Watershed connectivity | 12.3 | F |

## *Outcome sub-indices*

| Physical | Chemical | Biological | ECI |
|---|---|---|---|
| 0.75 | 0.6 | 0.77 | not assessed |

## *Metric measurements & curve indices*

| Function | Metric | Value | Index | Source | Curve (source) |
|---|---|---|---|---|---|
| Catchment hydrology | Agriculture, crop and hay | 3.0 | 0.97 | desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021) | EASI fixed criteria, Catchment hydrology (land-cover pressure) |
| Catchment hydrology | Impervious surface | 12.75 | 0.64 | desktop: StreamCat PCTIMP2019WS | EASI fixed criteria, Catchment hydrology (land-cover pressure) |
| Surface water storage | Wetland | 0.4 | ref. only | desktop: Computed (reference only) | StreamCurves reference-curve development |
| Reach inflow | Road density | 3.0 | 0.39 | desktop: STAF site engine v0.5.0 impervious (HR reach watershed, NLCD 2021) | EASI fixed criteria, Road-density proxy for concentrated inflow pressure |
| Streamflow regime | Base flow index | 12.75 | 0.18 | desktop: StreamCat PCTIMP2019WS | StreamCurves reference-curve development |
| Streamflow regime | Degree of regulation | 0.4 | 0.94 | desktop: Computed | EASI fixed criteria, Flow alteration (degree of regulation) |
| Low flow and baseflow dynamics | Bankfull width:depth ratio | 7.5 | 0.99 | field | StreamCurves reference-curve development |

| Function | Metric | Value | Index | Source | Curve (source) |
|---|---|---|---|---|---|
| Low flow and baseflow dynamics | Residual pool depth | | | | StreamCurves reference-curve development |
| Low flow and baseflow dynamics | Base flow index | 12.75 | 0.18 | desktop: StreamCat PCTIMP2019WS | StreamCurves reference-curve development |
| High flow dynamics | Impervious surface | 12.75 | 0.64 | desktop: StreamCat PCTIMP2019WS | EASI fixed criteria, Catchment hydrology (land-cover pressure) |
| Floodplain connectivity | Bankfull width:depth ratio | 7.5 | 0.99 | field | StreamCurves reference-curve development |
| Floodplain connectivity | Bankfull height above channel | | | | StreamCurves reference-curve development |
| Hyporheic connectivity | Sand + fines | | | field | StreamCurves reference-curve development |
| Hyporheic connectivity | Embeddedness | | | | StreamCurves reference-curve development |
| Channel evolution | Bankfull width:depth ratio | 7.5 | 0.99 | field | StreamCurves reference-curve development |
| Channel evolution | Bankfull height above channel | | | | StreamCurves reference-curve development |
| Channel and floodplain dynamics | Sinuosity | 12.5 | 1.00 | field | StreamCurves reference-curve development |
| Channel and floodplain dynamics | Bank angle | | | | StreamCurves reference-curve development |
| Sediment continuity | Relative bed stability | | | | StreamCurves reference-curve development |
| Sediment continuity | Sand + fines | | | field | StreamCurves reference-curve development |
| Bed composition and bedform dynamics | Substrate size | | | | StreamCurves reference-curve development |
| Bed composition and bedform dynamics | Large wood volume | | | | StreamCurves reference-curve development |
| Bed composition and bedform dynamics | Embeddedness | | | | StreamCurves reference-curve development |
| Light & thermal regime | Canopy density, mid-channel | 17.5 | 0.48 | field | StreamCurves reference-curve development |
| Carbon processing | Riparian veg cover, woody | | | | StreamCurves reference-curve development |
| Habitat provision | Fast-water / riffle habitat | | | | StreamCurves reference-curve development |
| Habitat provision | Natural fish cover | | | | StreamCurves reference-curve development |
| Watershed connectivity | Mapped dams within one mile | 0.4 | 0.82 | desktop: Computed | EASI fixed criteria, Nearby dam-proximity proxy |

### *Scoring advisories*

**Wetland**: engine-computed value shown as reference only: this curve was fitted on StreamCat predictors, so scoring it against an HR reach watershed value would mix training and serving sources

**Sinuosity**: value 12.5 is above the curve domain [1.0, 1.6272]; score clamped to the endpoint

## *Reference support*

Reference curves are built from least-disturbed stations chosen by a fixed landscape-pressure screen. Where this ecoregion has too few, comparable stations of its Level II and then its Level I ecoregion are used, and a metric with no defensible pool is not scored. Landscape pressure metrics are scored on fixed criteria.

| Metric | Scored against | Curve set |
|---|---|---|
| Agriculture, crop and hay | Fixed criteria, the same in every region | |
| Impervious surface | Fixed criteria, the same in every region | |
| Wetland | Reference curve from 10 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Road density | Fixed criteria, the same in every region | |
| Base flow index | Reference curve from 10 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Degree of regulation | Fixed criteria, the same in every region | |
| Bankfull width:depth ratio | Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Residual pool depth | Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Bankfull height above channel | Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Sand + fines | Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate | |
| Embeddedness | Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate | |
| Sinuosity | Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Bank angle | Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate | |
| Relative bed stability | Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Substrate size | Reference curve from 17 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk moderate | |
| Large wood volume | Reference curve from 14 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Canopy density, mid-channel | Reference curve from 12 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Riparian veg cover, woody | Reference curve from 14 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Natural fish cover | Reference curve from 14 least-disturbed stations borrowed from Level I ecoregion 8 (Eastern Temperate Forests), 0 of them in this ecoregion. Transfer risk high | |
| Mapped dams within one mile | Fixed criteria, the same in every region | |

## *Metrics not scored*

These metrics are not scored. A metric withheld for insufficient reference support has no curve. No pool of least-disturbed stations in this ecoregion or its parents supports it, no national donor pool was shown to transfer, no modeled expectation passed validation, and no published criterion applies.

| Metric | Functions | Reason |
|---|---|---|
| EPT taxa richness | Community dynamics | Insufficient reference support |
| Shannon diversity (H') | Community dynamics | Insufficient reference support |
| Tolerant individuals (%) ~ HBI | Community dynamics | Insufficient reference support |
| Benthic total taxa richness | Population support | Insufficient reference support |
| Chlorophyll a | Light and thermal regime | Insufficient reference support |

| Metric | Functions | Reason |
|---|---|---|
| Specific conductivity | Water and soil quality | Insufficient reference support |
| Dissolved nitrogen | Nutrient cycling | Insufficient reference support |
| pH | Water and soil quality | Insufficient reference support |
| Total phosphorus | Nutrient cycling | Insufficient reference support |
| Turbidity | Water and soil quality | Insufficient reference support |
| Native fish taxa richness | Population support | Insufficient reference support |

*Scores are computed automatically from the assessment's reference curves. Confirm state/region applicability of the curve source.*